

PANAMA CITY, FL, USA

WMO: 722245

Lat: 30.212N

Lon: 85.683W

Elev: 21

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 98-10

WBAN: 03882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.8	35.7	14.2	11.3	37.7	18.4	13.9	42.1	22.0	54.5	19.8	52.8	7.7	340	0.385

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	92.8	76.8	91.1	76.9	90.1	76.7	81.4	87.0	80.2	86.4	79.3	85.9	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.5	153.1	83.9	78.9	150.0	83.7	77.3	142.2	83.1	44.8	86.8	43.6	85.6	42.7	86.0	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.7	16.7	15.0	DB	26.1	97.6	3.3	2.4	23.7	99.3	21.7	100.7	19.9	102.1	17.5	103.8
			WB	22.4	83.4	3.1	1.8	20.1	84.7	18.3	85.7	16.6	86.7	14.3	88.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.4	54.0	56.0	62.1	68.2	75.7	81.1	82.8	82.5	79.5	71.8	62.6	55.7	(6)
	DBStd	12.20	9.51	7.41	6.68	5.60	4.06	2.71	2.59	2.64	3.78	6.83	7.33	8.61	(7)
	HDD50	148	74	26	4	0	0	0	0	0	0	0	5	39	(8)
	HDD65	1242	352	261	133	34	0	0	0	0	0	25	129	308	(9)
	CDD50	7227	197	194	379	547	796	933	1015	1008	885	675	381	217	(10)
	CDD65	2847	10	9	43	131	332	483	550	543	435	235	56	20	(11)
	CDH74	26464	10	13	134	617	2581	4895	6221	5928	4060	1697	273	35	(12)
	CDH80	9322	0	0	4	64	638	1792	2558	2374	1443	429	20	0	(13)
(14) Wind	WSAvg	6.8	7.5	7.0	7.4	7.3	7.2	6.2	5.8	5.5	7.3	6.7	6.7	7.3	(14)
(15) Precipitation	PrecAvg	53.2	4.4	4.4	5.2	3.7	2.3	4.6	5.7	6.8	5.6	3.3	3.5	3.7	(15)
	PrecMax	73.3	17.3	10.6	10.6	9.9	7.9	9.3	10.0	12.8	23.2	8.7	9.6	9.2	(16)
	PrecMin	36.7	1.4	0.8	0.0	0.8	0.0	0.6	1.8	2.8	0.1	0.6	0.5	0.8	(17)
	PrecStd	10.1	3.5	2.7	2.8	2.7	2.0	2.3	2.2	2.8	4.6	2.1	2.0	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	75.4	80.1	84.6	90.3	95.0	96.8	94.3	93.1	89.7	82.3	77.4	(19)
		MCWB	67.2	66.0	66.9	66.7	71.0	76.3	79.0	77.7	73.7	73.8	71.2	68.7	(20)
	2%	DB	72.9	72.9	77.3	81.9	87.8	91.5	92.5	91.6	90.7	87.2	80.7	74.6	(21)
		MCWB	66.8	65.0	65.6	68.8	71.6	76.1	77.8	78.2	74.2	74.0	70.6	68.3	(22)
	5%	DB	71.2	71.3	75.1	80.5	85.4	90.0	90.8	90.5	89.6	84.2	77.3	72.2	(23)
		MCWB	66.5	64.3	64.9	68.4	71.7	75.8	77.9	78.0	74.8	72.9	68.7	67.4	(24)
	10%	DB	68.5	68.4	72.9	78.3	83.7	88.1	89.8	89.7	87.8	82.0	74.8	70.1	(25)
		MCWB	64.1	62.7	64.6	67.0	72.1	75.4	77.8	77.8	75.2	71.7	67.3	66.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.4	70.3	72.5	75.0	78.1	80.9	84.0	82.8	80.2	79.7	76.1	72.1	(27)
		MCDB	72.3	72.6	75.8	80.6	84.1	88.3	88.5	88.2	85.1	83.6	80.1	74.8	(28)
	2%	WB	69.0	68.4	70.4	73.5	76.7	79.6	81.8	81.3	78.9	77.7	73.9	70.3	(29)
		MCDB	71.4	70.9	73.3	78.4	82.7	86.6	87.4	86.4	84.5	82.8	77.4	72.7	(30)
	5%	WB	66.9	66.5	68.7	72.1	75.7	78.4	80.3	80.2	78.1	76.0	71.9	68.6	(31)
		MCDB	69.8	68.9	72.5	76.4	81.7	85.3	86.9	86.2	84.4	81.5	75.3	71.1	(32)
	10%	WB	64.9	64.3	66.6	70.6	74.7	77.7	79.2	79.3	77.3	74.7	69.5	66.4	(33)
		MCDB	67.7	66.4	71.2	74.7	80.5	84.5	86.2	85.9	84.1	79.9	73.8	69.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	18.3	18.0	17.1	16.0	14.2	13.7	13.6	15.1	17.9	19.2	18.0	(35)
		MCDBR	16.2	18.3	18.3	17.9	17.6	16.2	15.3	15.1	16.9	17.6	19.1	16.9	(36)
		MCWBR	12.0	12.3	10.6	8.4	7.1	6.1	6.4	6.0	6.5	7.6	10.7	12.4	(37)
	5% WB	MCDBR	15.0	15.0	14.5	13.3	12.4	13.9	13.8	13.5	13.2	14.2	14.3	15.6	(38)
		MCWBR	12.9	13.0	10.8	8.0	6.5	5.9	7.2	6.4	6.1	7.8	10.1	13.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.335	0.349	0.368	0.395	0.437	0.468	0.498	0.493	0.436	0.384	0.355	0.343	(40)
	taud		2.536	2.499	2.450	2.375	2.290	2.260	2.191	2.202	2.338	2.465	2.482	2.526	(41)
	Ebn at Noon		285	291	291	285	272	262	254	254	266	275	275	276	(42)
	Edh at Noon		28	31	35	39	43	44	47	46	38	32	29	27	(43)
(44) All-Sky Solar Radiation	RadAvg		994	1198	1599	1923	2117	1959	1850	1723	1606	1435	1121	882	(44)
	RadStd		73	149	123	155	135	148	120	126	104	126	91	86	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-180	+301	+187

Nomenclature: See separate page